SECTION 4 · FINAL PROJECT DEMONSTRATION

8-bit Serial-In Parallel-Out (SIPO) Shift Register

Design, working and applications, simulated in Logisim

Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0
1	0	1	1	0	1	0	1

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Introduction and Objective

What is a SIPO shift register?

A shift register is a chain of flip-flops driven by a common clock. On every clock pulse, each bit moves one position along the chain. In a SIPO register, data enters bit by bit on a single serial line and is read out together on parallel outputs.

8	1	8
D flip-flops	serial input	parallel outputs

Objectives

1. Design an 8-bit SIPO shift register using eight D flip-flops.
2. Verify the shifting behaviour with a Logisim simulation.
3. Understand its working and real-world applications.

Theory and Logic Circuit

Basic concept

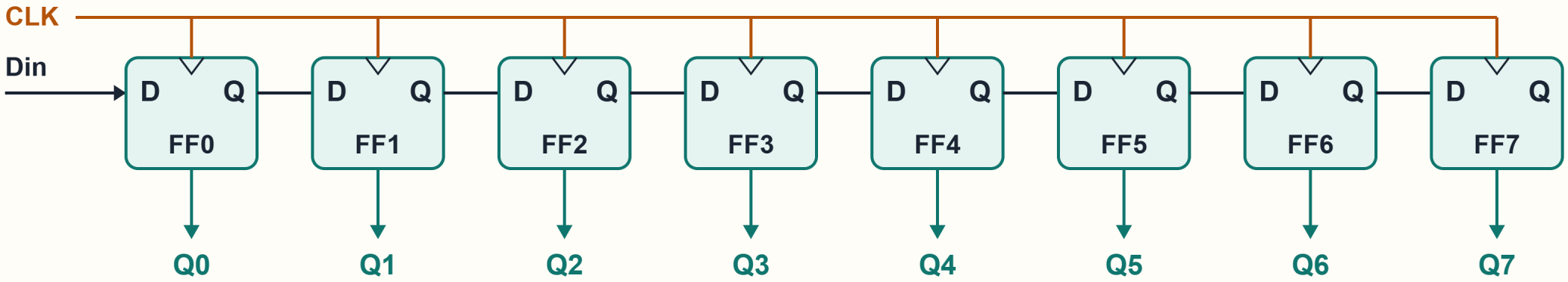
Data moves one position per clock edge. After 8 clocks, 8 serial bits are held in the register.

D flip-flop

Q copies D on the rising clock edge. Built from gated D latches (NAND gates) in master–slave form.

Connections

Din feeds the first D input, each Q feeds the next D, and one clock drives all stages.



Parallel outputs Q0 to Q7

Circuit Design in Logisim

How it works

1. Din connects to the D input of the first flip-flop.
2. On each clock pulse, every flip-flop copies the bit from the one before it.
3. All stored bits move one place along the chain.
4. After 8 pulses, Q7 to Q0 hold the complete byte.

Components: 8 D flip-flops, 1 input pin (Din), 1 clock, 8 output pins (LEDs).

Simulation Results

CLK	Din	Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0
1	1	0	0	0	0	0	0	0	1
2	0	0	0	0	0	0	0	1	0
3	1	0	0	0	0	0	1	0	1
4	1	0	0	0	0	1	0	1	1
5	0	0	0	0	1	0	1	1	0
6	1	0	0	1	0	1	1	0	1
7	0	0	1	0	1	1	0	1	0
8	1	1	0	1	1	0	1	0	1

Input stream, first bit first: 1 0 1 1 0 1 0 1. All outputs start cleared at 00000000.

After 8 clock pulses, Q7 to Q0 = 10110101, the same order as the input.

Conclusion and Applications

Outcome

- 8-bit SIPO built from 8 D flip-flops
- Verified in Logisim
- Serial data appears on Q7 to Q0 after 8 clocks
- Results match the theoretical shift behaviour

Applications

- Serial-to-parallel conversion in UART and SPI receivers
- Driving LEDs and displays with few pins
- I/O expansion for microcontrollers
- Data buffering in communication systems

Improvements

- Add clear and enable inputs
- Add parallel load for a universal register
- Cascade registers for 16 or 32 bits
- Add bidirectional shifting

Thank you. Questions?